

Real-Time Retargeting-Based Robust Polynomial Guidance for Chandrayaan-3 Lunar Landing Mission

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The main objective of the Chandrayaan-3 lunar landing mission was to achieve a safe and soft landing near the lunar south pole. The powered descent trajectory consisted of a braking phase with inertial navigation, an attitude hold, a braking phase with absolute navigation, and a terminal descent phase. This paper presents the flight-implemented closed-loop guidance design for the braking phase with absolute navigation. Our main contribution is the formulation of a robust and fuel-efficient analytical polynomial guidance algorithm under state and control constraints with real-time retargeting capabilities. This is achieved through offline data-driven analysis, which enables a robust online decision-making algorithm to assess the feasibility of pinpoint landing in real time at the start of the phase. If the pinpoint landing is deemed infeasible, the algorithm computes a suitable alternative retargeting site in real time without the need for onboard optimization. Consequently, the proposed algorithm is analytical and extends the operational region of the classical polynomial guidance algorithm.

Nomenclature

a	=	acceleration of lander, m/s ²
(c, b)	=	coefficients of decision boundary
$\mathcal{D}$	=	t_{go} dataset
g	=	moon gravity vector, m/s ²
H	=	altitude, km
(h, k, θ)	=	centroid and rotation angle of conic
I_{sp}	=	specific impulse for engine, s
K	=	t_{go} coefficients
m	=	mass of lander, kg
$\mathbb{P}(x, y)$	=	projection function
$\mathbf{r}$	=	position vector of lander, km
S	=	downrange, km
$\mathcal{S}_T^i$	=	feasible t_{go} dataset for i th data
T	=	thrust from engine, N
$\mathcal{T}$	=	guidance dataset
$\mathcal{T}_c$	=	feasible dataset
$\mathcal{T}_{nc}$	=	infeasible dataset
v	=	horizontal velocity, m/s
$\mathbf{v}$	=	velocity vector of lander, m/s
w	=	vertical velocity, m/s
$\mathcal{X}$	=	perturbation dataset
(x, y)	=	reduced state space
Z	=	feature vector
Δ_T	=	maximum thrust rate realizable from engine, N/s
$(\delta_h, \delta_y, \delta_\theta)$	=	shift in conic parameters
θ_{awv}	=	cone angle with respect to vertical, deg

I. Introduction

ON August 23, 2023, India achieved a major milestone in space exploration by successfully performing a soft landing near the south pole region of the moon. Chandrayaan-3 was configured as a two-module stack comprising a lander and a propulsion module. The GSLV MKIII launch vehicle injected the combined module into an Earth elliptical parking orbit (EEPO) of 180 km $\times$ 36,000 km. From the EEPO, a series of orbit maneuvers were carried out to raise the apogee altitude and eventually perform the trans-lunar insertion (TLI). These maneuvers were performed using the propulsion module. Subsequently, lunar orbit maneuvers were performed to achieve a low lunar orbit (LLO) of 100 km $\times$ 100 km. The lander module was separated from the propulsion module, and a deorbit maneuver was performed to achieve a powered descent with initial conditions. The powered descent orbit was chosen as 30 km $\times$ 100 km based on ground trajectory optimization studies and orbit stability. Landing site characterization was done a priori, and a landing strip of nearly 4 km along track and 2.4 km across track was chosen to meet associated engineering constraints such as local surface undulation, average slope, boulder sizes, etc. More details on landing site characteristics can be found in [1,2]. Within this region, a designated landing point was identified as the guidance target, selected based on the largest neighboring hazard-free zone.

Powered descent guidance technology plays a key role in generating feasible reference trajectories onboard to ensure a safe and soft landing. In the context of lunar landing missions like Chandrayaan-3, where navigation errors and actuation dispersions can significantly impact the feasibility of achieving pinpoint landing, guidance algorithms must be both robust and adaptive. A critical capability of the guidance system is to assess, in real time, whether a soft landing at the predesignated target is feasible and, if not, to autonomously retarget to an alternate site within the allowable landing strip. Furthermore, the reliability and complexity of the onboard computation are essential for practical implementations.

The powered descent guidance problem for pinpoint landing is typically formulated as an optimal control problem with various state and control constraints. Over the last two decades, numerous analytical and numerical approaches have been reported in the literature for solving the pinpoint and soft landing problem. Trajectory optimization methods currently play a major role in generating fuel-optimal trajectories, with the potential for real-time onboard implementation [3]. Legendre pseudo-spectral methods are used to develop numerical solutions for the optimal control problem in [4]. A unified framework for solving multiphase optimal control problems using pseudo-spectral methods is developed in [5]. These numerical methods rely on transcribing the infinite-dimensional

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control problem to a finite-dimensional discrete state-action control problem, which can be solved using nonlinear programming solvers [6–8]. While such methods are useful in mission planning, the iterative nature of a general nonlinear program limits their applicability in onboard implementation. In [9–11], minimum-fuel powered descent guidance for pinpoint landing has been developed using lossless convexification, which is well suited for real-time applications. The resulting convex program is formulated as a second-order cone program (SOCP) that can be solved with numerically efficient interior-point solvers that have formal convergence guarantees and deterministic stopping criteria in bounded polynomial time. This approach is extended to compute the minimum landing error trajectory when no feasible trajectory to the target exists in [12]. Direct trajectory optimization methods have the advantage of allowing explicit enforcement of control and state constraints. However, the Chandrayaan-3 onboard system employs a single-processor architecture based on the UT699 LEON-3 processor (48 MHz), which executes the GNC algorithms along with other subsystems such as thermal, power, telemetry, and telecommand. Consequently, the computational resources available for guidance are limited. A similar constraint also motivated the use of analytical guidance in the SLIM lander [13].

Another popular class of indirect methods used to solve the guidance problem utilizes the application of the Pontryagin maximum principle (PMP) to solve the two-point boundary value problem that arises from the necessary conditions of optimality. In [14,15], a three-dimensional fuel-optimal powered descent guidance algorithm is presented based on indirect methods. While these methods can handle control constraints such as thrust magnitude limits and thrust-pointing constraints, they typically require a good initial guess for smooth convergence, and incorporating state constraints remains challenging.

Several analytical approaches to the guidance problem have also been reported, beginning with the Apollo guidance framework. In [16,17], the analytical optimal feedback guidance algorithm is derived using the calculus of variations under a reasonable set of approximations in the cost function, state-control constraints, and dynamics. The Apollo-type polynomial guidance algorithm based on linear and quadratic polynomials in time also results in an analytical closed-loop solution [18,19]. Inspired by the Apollo polynomial guidance framework, a generalized polynomial solution based on arbitrary order was developed in [20]. It was demonstrated that the performance of polynomial guidance could be improved by increasing the number of free parameters, specifically by increasing the polynomial order in the acceleration profile, without altering the number of boundary conditions. The optimal order is then selected by minimizing a suitable performance metric, such as control effort. A number of higher-order polynomial guidance algorithms were tested, and it was determined that a cubic polynomial was sufficient. Recently, the authors in [21] demonstrated, using optimal control theory, that cubic polynomial guidance is the optimal solution to the minimum-jerk problem in the absence of state and control constraints. A family of tunable guidance laws is proposed in [22] in a unifying framework that offers the flexibility to balance the competing considerations among fuel usage, trajectory shaping, and guidance command variations. Polynomial guidance is attractive due to its analytical simplicity and accuracy; guidance implementation is straightforward, and it can theoretically drive the spacecraft to the target. However, despite its appeal, the algorithm is generally unable to enforce thrust or flight constraints.

A significant advancement toward analytical fuel-optimal guidance was presented in [13], where the authors proposed a fuel-optimal analytical guidance law for the terminal descent phase of lunar landing. Their key finding, derived from numerically solving the fuel-optimal powered descent guidance problem, is that during the terminal descent phase—when the module begins with nearly zero horizontal velocity and is required to travel a horizontal distance to a safe landing site based on navigation input—the phase can be modeled as a minimum-thrusting segment (free fall, if feasible, as done in the SLIM lander [23]) to build vertical velocity using gravity, perform horizontal position correction, and then execute

vertical braking. This formulation enables analytical computation of the optimal transition point to initiate horizontal braking, as the thrust profile involves only a single switching surface. However, in the absolute navigation phase of Chandrayaan-3, where the vehicle possesses significant horizontal and vertical velocities and a substantial downrange to cover, such phase decomposition within the guidance sequence is not feasible due to the presence of multiple switching surfaces in the bang–bang thrust profile [24].

Recent trends include learning-based guidance. In [25], adaptive polynomial guidance is proposed, and guidance gains are learned using deep reinforcement learning techniques. In [26,27], a deep-reinforcement-learning-based approach is developed for fuel-optimal powered descent guidance. In [28], the authors compared unconstrained and constrained optimization techniques for formulating a convex decision-boundary guidance feasibility check by considering only the boundary points separating the feasible zone from the infeasible zone. A successful integration of supervised learning into guidance design was also demonstrated in the SLIM lander [23] during its powered descent phase, which consisted of three boost phases and two coast phases. In this implementation, supervised learning was employed to develop a predictive module that determined the start time of each boost phase, enabling effective phase transitions during descent.

This paper presents the flight-implemented powered descent guidance used during the absolute navigation phase of Chandrayaan-3. The main contributions of this work are twofold: first, the development of a robust and fuel-efficient analytical polynomial guidance law that incorporates control and state constraints through an offline data-driven analysis, extending the classical polynomial guidance approach and, second, the development of a robust real-time decision-making algorithm that evaluates the feasibility of a pinpoint landing at the start of the guidance phase. If the pinpoint landing is deemed infeasible, the algorithm provides suitable alternative retargeting sites in real time without solving any onboard optimization. Although data-driven optimization and other similar techniques have previously been applied to aerospace guidance and control problems, this appears, to the best of the authors' knowledge, to be the first successful implementation of data-driven augmentation to polynomial guidance that supports real-time retargeting, is analytical, and maximizes the operational region of guidance for soft landing.

The remainder of this paper is organized as follows. Section II provides an overview of the Chandrayaan-3 powered descent and landing trajectory. Section III presents the guidance formulation, including functional requirements and associated constraints for the absolute navigation phase of the mission. Section IV presents the cubic polynomial guidance algorithm. Section V presents the analytical time-of-flight prediction model based on offline data-driven analysis. Section VI presents the real-time retargeting algorithm. Section VII presents extensive simulation results, including a comparison with fuel-optimal powered descent guidance, Monte Carlo analysis, and evaluation of the guidance operating region by the retargeting algorithm. Section VIII concludes the paper.

II. Overview of Chandrayaan-3 Powered Descent Sequence

At perilune altitude of 30 km, powered descent was initiated using an autonomous closed-loop guidance, navigation, and control (GNC) system. The powered descent phase was composed of multiple subphases to ensure a safe and soft landing. The powered descent subphases were composed of an inertial navigation phase, an attitude hold phase, an absolute navigation phase, and a terminal descent phase.[†] The terminal descent phase consists of hovering at a higher altitude, vertical descent, and hazard avoidance, followed by constant-velocity descent. The entire powered descent profile for Chandrayaan-3 lasted for approximately 1140 s, ensuring a soft

[†]In other publications on Chandrayaan-3, inertial and absolute navigation phases are also referred to as rough braking and fine braking, respectively, where “rough” and “fine” denote the respective levels of navigation accuracy.

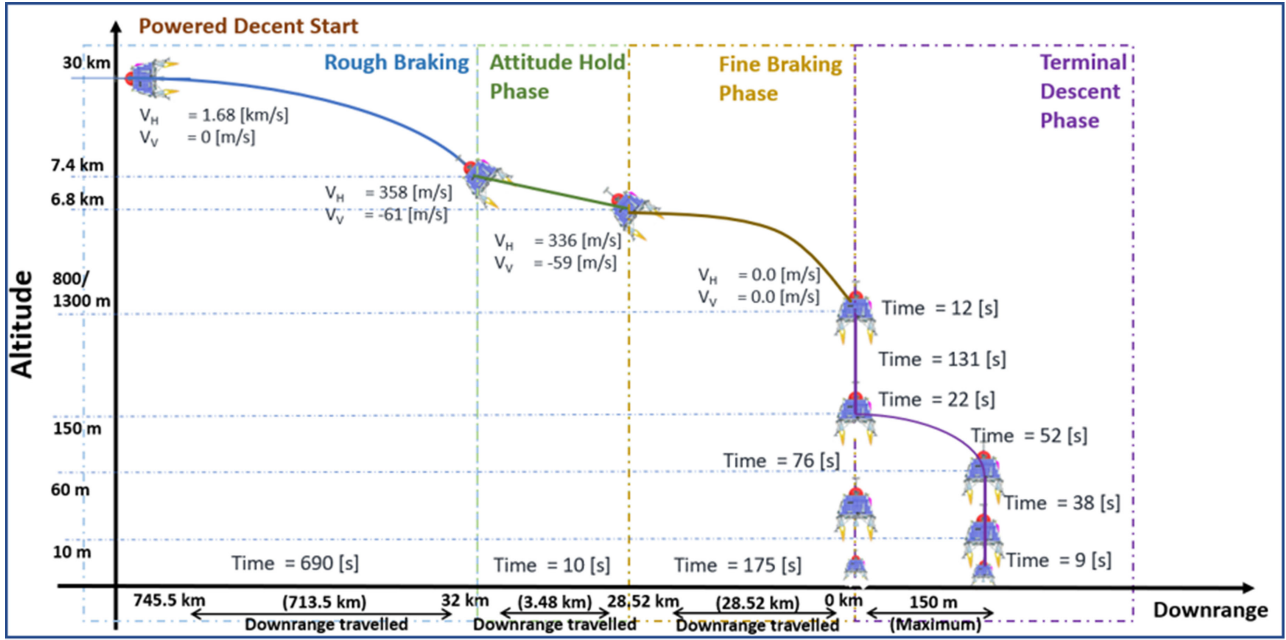


Fig. 1 Powered descent phases of Chandrayaan-3 [1]. Here, rough braking refers to braking phase with inertial navigation, and fine braking refers to braking phase with absolute navigation.

landing on the moon. Figure 1 shows the powered descent and landing trajectory profile for Chandrayaan-3.

Each of the powered descent phases was designed with a specific objective. The inertial navigation phase begins at 30 km altitude and terminates at 7.4 km. This phase is aimed at reducing velocity significantly and covering the maximum downrange distance to the target using inertial navigation for state estimation. The attitude hold phase was designed to obtain absolute sensor updates and aid inertial navigation. The absolute navigation phase aimed to reduce velocity to zero, align the module vertically, and bring it to a safe height of 800 m above the designated landing site to initiate terrain-relative navigation. Navigation in this phase relied on inertial sensors with periodic updates from absolute sensors. The terminal descent phase is further split into multiple phases to carry out terrain-relative navigation, hazard detection, and avoidance for a safe landing. More details on the landing profile and powered descent trajectory design can be found in [1,29,30].

The initial conditions of this phase can vary significantly, mainly due to inertial navigation drift in earlier phases, residual guidance error, and thrust and mass perturbations. The targeted landing site is selected before launch; however, large deviations in the initial conditions of this phase may render pinpoint landing at the designated site infeasible. Therefore, a feasibility check is performed at the start of the absolute navigation phase. If pinpoint landing is not possible, the guidance algorithm retargets to an alternate location within the predefined landing strip with minimal deviation from the preselected landing point.

III. Problem Formulation

The guidance and control implementation of Chandrayaan-3 follows a hierarchical approach. The outer guidance loop generates steering commands, determining both the required thrust and the reference orientation of the lander to guide it from its current state to the terminal state. The inner attitude control loop tracks the reference orientation, while the propulsion system regulates thrust based on measured acceleration from the onboard accelerometer to meet the required thrust demand. The inner-loop attitude controller operates at a much higher bandwidth compared to the outer-loop guidance time horizon. This allows the translational dynamics of the lander to be decoupled from its attitude dynamics for closed-loop guidance design. In this work, we consider the decoupled translational dynamics for guidance design.

A. Governing Dynamics

The lander translational dynamics are considered under the non-rotating flat moon approximation. Due to the significant velocity reduction in the inertial navigation phase, the Coriolis and moon's rotational velocity terms in the dynamics can be neglected, as their contribution to the total acceleration is negligible. Under such assumptions, the lander translational dynamics are given as

$$\begin{aligned}\dot{\mathbf{r}} &= \mathbf{v} \\ \dot{\mathbf{v}} &= \mathbf{g} + \mathbf{a} \\ \|\mathbf{a}\| &= \frac{T}{m} \\ \dot{m} &= -\frac{T}{I_{sp}g_0}\end{aligned}\quad (1)$$

where $\mathbf{r} \in \mathbb{R}^3$, $\mathbf{v} \in \mathbb{R}^3$, and $\mathbf{a} \in \mathbb{R}^3$ are the position, velocity, and acceleration vectors, respectively; T is the total thrust from the propulsion system; m is the mass of the lander; $\mathbf{g}$ is the moon gravity vector; I_{sp} is the specific impulse of the engine; and g_0 is the constant associated with the fuel consumption rate and equals $9.81 \text{ m} \cdot \text{s}^{-2}$.

B. Guidance Functional Requirement

The requirements for the absolute navigation phase of Chandrayaan-3 are derived from the mission constraints. The guidance module should steer the lander from the current state to the terminal state while satisfying both initial and terminal acceleration constraints. Additionally, the guidance module should compute acceleration in a closed loop, ensuring fuel efficiency and robustness to state and propulsion dispersions. If pinpoint landing at the designated site is not feasible, the system should detect it in real time at the beginning of the phase. In such cases, the guidance algorithm should autonomously retarget to a safe landing site.

C. Guidance Constraints

As per the guidance functional requirement (GFR), the boundary conditions are expressed as

$$\begin{aligned}\mathbf{r}(t_0) &= \mathbf{r}_0, & \mathbf{r}(t_f) &= \mathbf{r}_f \\ \mathbf{v}(t_0) &= \mathbf{v}_0, & \mathbf{v}(t_f) &= \mathbf{v}_f \\ \mathbf{a}(t_0) &= \mathbf{a}_0, & \mathbf{a}(t_f) &= \mathbf{a}_f\end{aligned}\quad (2)$$

where $\mathbf{r}_0$, $\mathbf{v}_0$, and $\mathbf{a}_0$ are the initial values of the position, velocity, and acceleration vectors, and $\mathbf{r}_f$, $\mathbf{v}_f$, and $\mathbf{a}_f$ are the terminal values of the position, velocity, and acceleration vectors.

The Chandrayaan-3 lander is equipped with four throttleable liquid engines to perform braking operations. Each engine operates within the operating range of 370–780 N. The throttling rate $\dot{T}$ depends on the operating thrust and is upper bounded by the hardware capabilities. The associated control constraints are given by

$$0 \leq T_{\min} \leq T \leq T_{\max} \quad (3)$$

$$\dot{T} \leq \Delta_T \quad (4)$$

where $T_{\min}$ and $T_{\max}$ denote the minimum and maximum operating thrust from each engine, and Δ_T denotes the upper bound on the rate of change in thrust. Equations (3) and (4) are associated with the physical limits on the thrust and thrust-rate constraints imposed by the engine, respectively.

It is required that the trajectory does not go below the target during the maneuver. Additionally, the thrust vector's deviation from the vertical is constrained within a specified tolerance, dictated by sensor operations such as altimeters and Doppler velocity meters. The respective state constraints are given by

$$r_1(t) \geq r_f(1) \quad \forall t \in [0, t_f] \quad (5)$$

$$|\theta_{\text{avv}}| \leq \theta_{\text{tol}} \quad (6)$$

where r_1 is the height component of the current position vector, $r_f(1)$ is the height component of the terminal position vector, θ_{avv} is the thruster vector cone angle with respect to the vertical, with a cone apex at the target landing site, and θ_{tol} is the tolerance in cone angle.

IV. Polynomial Guidance Formulation

Inspired by the Apollo polynomial guidance framework, a generalized polynomial solution based on arbitrary-order polynomials was developed in [20]. It was demonstrated that the performance of polynomial guidance could be improved by increasing the number of free parameters, specifically by increasing the polynomial order in the acceleration profile, without altering the number of boundary conditions. The optimal order is then selected by minimizing a suitable performance metric, such as control effort. A number of higher-order polynomial guidance algorithms were tested, and it was determined that a cubic polynomial was sufficient. Philosophically, our approach is similar; however, it differs in that we introduce different free parameters for optimization while keeping the polynomial order fixed.

We first address the feasibility problem of determining a feedback guidance solution that satisfies the system dynamics and boundary conditions. This is achieved by prescribing a polynomial of sufficient order and computing the coefficients to meet these constraints. Among the resulting family of feasible solutions, a fuel-efficient solution is selected by introducing a free parameter for optimization. In our approach, this free variable is the time-to-go ($t_{\text{go}} = t_f - t$), which represents the remaining time to reach the target. This section summarizes the formulation of polynomial guidance. The details of parameter optimization for t_{go} are deferred to Sec. V.

A. Cubic Polynomial Guidance Formulation

The lander's boundary conditions are specified in terms of position, velocity, and acceleration constraints to maintain phase continuity. Therefore, we represent the net guidance acceleration, $\bar{\mathbf{a}}(t) = \mathbf{a}(t) + \mathbf{g}$, using a third-order polynomial as

$$\bar{\mathbf{a}}(t) = \mathbf{C}_0 + \mathbf{C}_1 t + \mathbf{C}_2 t^2 + \mathbf{C}_3 t^3 \quad (7)$$

Integrating the above acceleration profile leads to

$$\mathbf{v}(t) = \mathbf{v}_0 + \mathbf{C}_0 t + \mathbf{C}_1 \frac{t^2}{2} + \mathbf{C}_2 \frac{t^3}{3} + \mathbf{C}_3 \frac{t^4}{4} \quad (8)$$

$$\mathbf{r}(t) = \mathbf{r}_0 + \mathbf{v}_0 t + \mathbf{C}_0 \frac{t^2}{2} + \mathbf{C}_1 \frac{t^3}{6} + \mathbf{C}_2 \frac{t^4}{12} + \mathbf{C}_3 \frac{t^5}{20} \quad (9)$$

The polynomial coefficients are obtained by enforcing the dynamics and boundary conditions through a series of linear systems of equations:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & t_{\text{go}} & t_{\text{go}}^2 & t_{\text{go}}^3 \\ t_{\text{go}} & \frac{t_{\text{go}}^2}{2} & \frac{t_{\text{go}}^3}{3} & \frac{t_{\text{go}}^4}{4} \\ \frac{t_{\text{go}}^2}{2} & \frac{t_{\text{go}}^3}{6} & \frac{t_{\text{go}}^4}{12} & \frac{t_{\text{go}}^5}{20} \end{bmatrix} \begin{bmatrix} \mathbf{C}_0^i \\ \mathbf{C}_1^i \\ \mathbf{C}_2^i \\ \mathbf{C}_3^i \end{bmatrix} = \begin{bmatrix} \bar{\mathbf{a}}_0^i \\ \bar{\mathbf{a}}_f^i \\ \mathbf{v}_f^i - \mathbf{v}_0^i \\ \mathbf{r}_f^i - \mathbf{r}_0^i - \mathbf{v}_0^i \cdot t_{\text{go}} \end{bmatrix} \quad (10)$$

where $\mathbf{r}_0$, $\mathbf{v}_0$, and $\bar{\mathbf{a}}_0$ represent the initial position, velocity, and acceleration, respectively, and $\mathbf{r}_f$, $\mathbf{v}_f$, and $\bar{\mathbf{a}}_f$ represent the terminal position, velocity, and acceleration, respectively. Index i represents the indices of the x , y , z components of the vector.

In each guidance cycle, the net acceleration is computed by substituting the current states of the lander as the initial states to determine the polynomial coefficients. The guidance acceleration for the current cycle is then obtained as

$$\bar{\mathbf{a}} = \mathbf{C}_0 + \mathbf{C}_1 g_{\text{step}} + \mathbf{C}_2 g_{\text{step}}^2 + \mathbf{C}_3 g_{\text{step}}^3 \quad (11)$$

where g_{step} is the guidance cycle time.

The total effective acceleration that the lander must generate using the onboard propulsion system is computed by subtracting the lunar gravity from the net guidance acceleration. As t_{go} approaches zero, the guidance computation approaches singularity. To prevent this, the computation of guidance coefficients is terminated once t_{go} reaches a predefined threshold. However, the execution of the guidance continues until t_{go} reaches zero using the last computed coefficients. This strategy helps in minimizing terminal errors.

V. Formulation of t_{go} via Data-Driven Analysis

Several approaches for computing t_{go} have been reported in the literature, beginning with the Apollo guidance framework. These methods typically rely on simplifying assumptions applied either to system constraints or the trajectory itself. For example, in the Apollo guidance law [18,31], t_{go} is selected such that the vertical component of the acceleration varies linearly with time. In the optimal feedback guidance framework [16], t_{go} is derived from transversality conditions obtained through the application of the calculus of variations to a simplified optimal control problem that neglects thrust constraints and mass flow rate dynamics. Similarly, in the minimum-jerk formulation [21], t_{go} is computed online considering acceleration minimization as a secondary control objective. A major limitation of the Apollo guidance approach is that it restricts the trajectory space by enforcing a linear profile on the vertical component of acceleration. Similarly, optimal feedback guidance methods ignore state and actuation constraints, limiting their applicability in constrained scenarios. Therefore, we propose treating t_{go} as a free variable in offline parametric analysis in a supervised learning framework, allowing robustness and suboptimality to be incorporated into the guidance design.

We now describe the supervised learning pipeline for data generation and training. A large dataset is generated by sampling the input space of the guidance algorithm, accounting for various sources of perturbations and their bounded uncertainties. This is done using high-fidelity six-degree-of-freedom (6DoF) simulation software developed for end-to-end mission simulation and analysis. The 6DoF simulator models full system dynamics, including the physical behavior and dynamic characteristics of all sensing and

A 4D mesh of a perturbation matrix with dispersion bounds of ± 3 km in downrange and altitude in steps of 200 m and ± 17 m/s in horizontal and vertical velocity in steps of 1 m/s is generated. The resultant perturbation dataset $\mathcal{X}$ is given as

	Nominal	Additive 3σ error	Dispersion bound for simulation
/s	6.8	± 0.9	± 3.0
	28.5	± 0.75	± 3.0
	59	± 8	± 17
	336	± 6	± 17
	780	± 48	± 48

$$\mathcal{S}_T^i := \{(t_{\text{go},j}; m_{f,j}), j = 1, 2, \dots, K_i \mid i = 1, 2, \dots, n\} \quad (13)$$

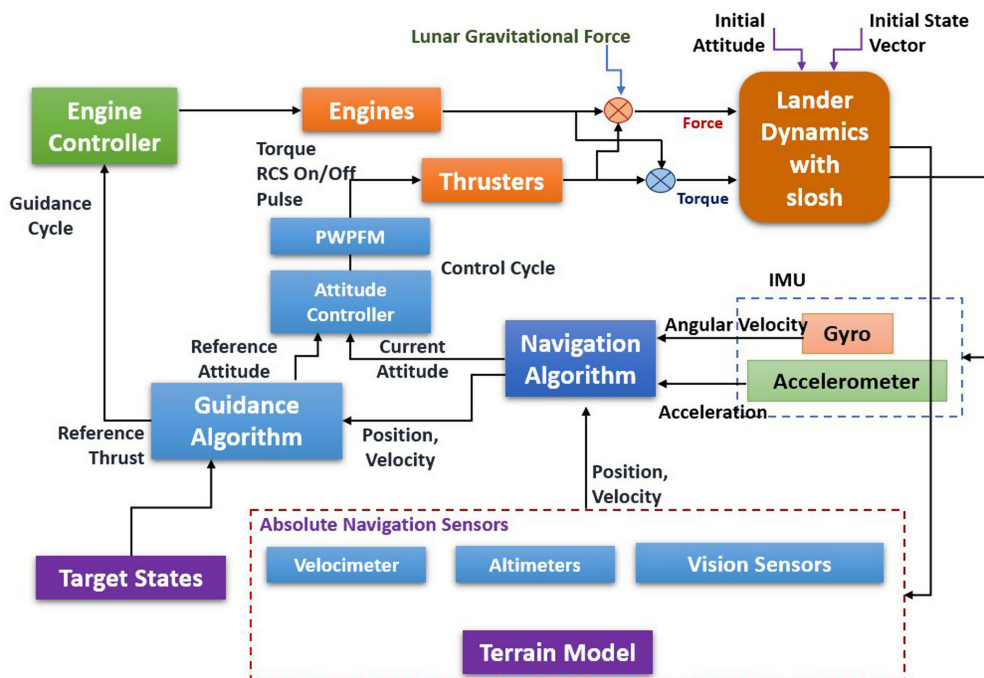
$$t_{\text{go},i}^* = \arg \max_{(t_{\text{go}}, m_f) \in \mathcal{S}_T^i} m_f \quad (14)$$


Fig. 2 Six-DoF simulation software block diagram.

This ensures the selection of a fuel-optimal solution for each perturbed initial state. It is obtained trivially once the dataset $\mathcal{S}_T^i$ is available.

Finally, we obtain the supervised learning dataset $\mathcal{D}$ as

$$\mathcal{D} = \{(\mathcal{X}_i, t_{go,i}^*) | \mathcal{X}_i \in \mathbb{R}^4; t_{go,i} \in \mathbb{R}, i = 1, 2, \dots, n\} \quad (15)$$

Algorithm 1 details the dataset generation procedure. Mathematically, $t_{go}^*(H, S, w, v)$ is then expressed as

$$t_{go}^* = K^T \phi(H, S, w, v; \mathcal{D}) \quad (16)$$

where ϕ defines the function approximator used for fitting the data. We choose ϕ to be a polynomial basis up to order 2. The coefficient vector K is obtained using Lasso regression [33] on dataset $\mathcal{D}$ as

$$\min_K \left(\frac{1}{2} \sum_{i=1}^{|\mathcal{D}|} (t_{go,i} - K^T \phi(\mathcal{X}_i))^2 + \mu \|K\|_1 \right) \quad (17)$$

where $|\mathcal{D}|$ is the cardinality of set $\mathcal{D}$, $\mu \|K\|_1$ is the regularization term, $\|K\|_1$ denotes the l_1 -norm of the coefficient vector K , and μ governs the tradeoff between model fidelity and complexity. In particular, it controls the strength of the ℓ_1 -norm penalty applied to the regression coefficients, promoting sparsity in the resulting model. This sparsity is desirable in our context, as it yields a simplified analytical expression for t_{go} , making it suitable for onboard implementation. A higher value of μ forces more coefficients to zero, potentially improving interpretability and reducing

computational overhead. However, an excessively large value may lead to underfitting by discarding important features. In contrast, a very small μ may retain all features, leading to overfitting and reduced generalization [33–35].

VI. Real-Time Retargeting via Convex Optimization

To enable autonomous real-time retargeting under off-nominal scenarios, we present a zero-loss classifier that distinguishes initial states in the controllable region from those in the uncontrollable region. To characterize the operational region over which the guidance framework discussed in the previous section remains valid, we performed extensive 6DoF simulations for each initial condition in the dataset $\mathcal{X}$. In this context, the operational region is defined as the set of initial conditions for which the guidance law drives the lander states to the desired terminal states within specified tolerance limits. We use cubic polynomial guidance with t_{go} coefficients predicted via Lasso regression to construct the guidance dataset $\mathcal{T}$:

$$\mathcal{T} = \{(\mathcal{X}_i, d_i) | \mathcal{X}_i \in \mathbb{R}^4, d_i \in \{-1, +1\}, i = 1, 2, \dots, n\} \quad (18)$$

where d_i takes $+1$ when represents the controllable region and -1 when represents the uncontrollable region.

Figure 3 shows the dataset $\mathcal{T}$ projected in 2D using dimensional reduction by considering the ratio of position and velocity. The nominal operating condition, marked as a “diamond,” lies well within the controllable zone. The green region represents the set of initial conditions for which the guidance algorithm drives the lander states to the desired terminal states and is referred to as the controllable set,

Algorithm 1: Algorithm for dataset generation for t_{go}

1:	Input: Dataset $\mathcal{X}$, $t_{go}^{\min}$, $t_{go}^{\max}$, hyperparameters ϵ_1 , ϵ_2 , ϵ_T , δt_{go}	
2:	Output: Dataset $\mathcal{D} = \{(\mathcal{X}_i, t_{go,i}^*)\}$	
3:	Define convergence indicator function, $\mathcal{I}_c(\mathcal{X}_i, t_{go})$, for 6DoF simulation run:	
	$\mathcal{I}_c = \begin{cases} 1, & \text{if } \ \mathbf{r}(t_f) - \mathbf{r}_f\ < \epsilon_1, \quad \ \mathbf{v}(t_f) - \mathbf{v}_f\ < \epsilon_2, \quad r_1(t) \geq r_f(1), \theta_{awv} \leq \theta_{tol}, \quad T \in [T_{\min}, T_{\max}], \dot{T} \leq \Delta_T \\ 0, & \text{otherwise} \end{cases}$	
4:	for $i \in \mathcal{X}$ do	
5:	Initialize: $\mathcal{S}_T^i \leftarrow \emptyset$, $g_{\text{conv}} \leftarrow 0$, $t_{go,i}^{\min} \leftarrow t_{go}^{\min}$, $t_{go,i}^{\max} \leftarrow t_{go}^{\max}$	
6:	while $(t_{go,i}^{\max} - t_{go,i}^{\min}) > \epsilon_T$ do	▷ computation of minimal feasible t_{go}
7:	$t_{go}^{\text{current}} \leftarrow \frac{t_{go,i}^{\min} + t_{go,i}^{\max}}{2}$	
8:	Run 6DoF simulation with $(\mathcal{X}_i, t_{go}^{\text{current}})$ and evaluate $\mathcal{I}_c$	
9:	if $\mathcal{I}_c == 1$ then	
10:	$g_{\text{conv}} \leftarrow 1$, $t_{go,i}^{\max} \leftarrow t_{go}^{\text{current}}$	
11:	else	
12:	$t_{go,i}^{\min} \leftarrow t_{go}^{\text{current}}$	
13:	end if	
14:	end while	
15:	Record $(t_{go,i}^{\min}, m_{f,i})$ in $\mathcal{S}_T^i$	
16:	Initialize: $t_{go,i}^{\max} \leftarrow t_{go}^{\max}$	
17:	while $(t_{go,i}^{\max} - t_{go,i}^{\min}) > \epsilon_T$ do	▷ Construction of set $\mathcal{S}_T^i$
18:	Run 6DoF simulation with $(\mathcal{X}_i, t_{go,i}^{\max})$ and evaluate $\mathcal{I}_c$	
19:	if $\mathcal{I}_c == 1$ then	
20:	Record $(t_{go,i}^{\max}, m_{f,i})$ in $\mathcal{S}_T^i$	
21:	else	
22:	continue	
23:	end if	
24:	Update $t_{go,i}^{\max} \leftarrow t_{go,i}^{\max} - \delta t_{go}$	
25:	end while	
26:	if $\mathcal{S}_T^i \neq \emptyset$ then	
27:	Select $t_{go,i}^* = \arg \max_{(t_{go}, m_f) \in \mathcal{S}_T^i} m_f$	
28:	$\mathcal{D} \leftarrow \mathcal{D} \cup \{(\mathcal{X}_i, t_{go,i}^*)\}$	▷ Construction of set $\mathcal{D}$
29:	end if	
30:	end for	
31:	return $\mathcal{D}$	

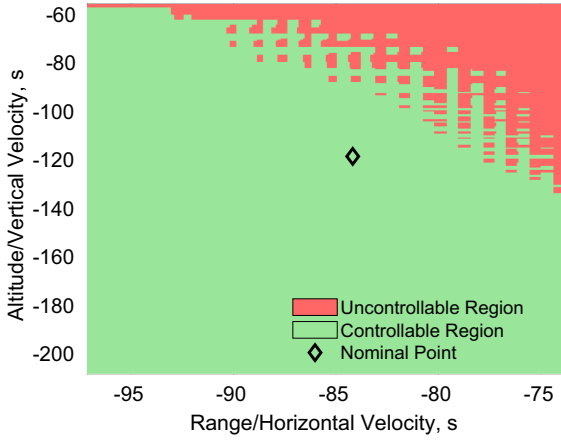


Fig. 3 Guidance dataset.

$\mathcal{T}_c$. Conversely, the red region is referred to as the uncontrollable set, $\mathcal{T}_{nc}$. These sets are formally defined as

$$\begin{aligned}\mathcal{T}_c &:= \{(\mathcal{X}_i, d_i) \in \mathcal{T} \mid d_i = +1\} \\ \mathcal{T}_{nc} &:= \{(\mathcal{X}_i, d_i) \in \mathcal{T} \mid d_i = -1\}\end{aligned}\quad (19)$$

We aim to enhance the robustness of the guidance system to larger state perturbations by expanding the controllable set $\mathcal{T}_c$. Previously, robustness was introduced by treating t_{go} as a free variable in the optimization. However, beyond a certain region of the state space, no feasible t_{go} exists for cubic polynomial guidance. In such cases, while pinpoint landing may not be achievable, a soft landing remains possible, thanks to the $4 \text{ km} \times 2.4 \text{ km}$ landing ellipse available from Optical High-Resolution Camera (OHRC) imaging by Chandrayaan-2. To achieve this, we propose relaxing the range as a free parameter during offline analysis, enabling a graceful degradation from pinpoint landing to soft landing. This is accomplished through a robust, real-time decision-making algorithm that assesses the feasibility of pinpoint landing at the start of the phase. If pinpoint landing is deemed infeasible, the algorithm provides a suitable alternative retargeting site. We refer to this novel supervisory layer, integrated with polynomial guidance, as real-time retargeting-based polynomial guidance.

A. Convex Decision Boundary Formulation

Consider the guidance dataset ($\mathcal{T}$) shown in Fig. 3. This dataset can be effectively modeled using a function approximator suitable for onboard evaluation using real-time state parameters. A convex function is well suited for this purpose, as it naturally distinguishes between controllable and uncontrollable regions: it evaluates to one sign inside the convex region and the opposite sign outside, while evaluating to zero on the boundary. This property enables reliable real-time decision-making.

Consider a conic function to represent the convex decision boundary in a reduced-dimensional state space. Let us define the reduced state space as $x = S/v$ and $y = H/w$. Then, the conic function representing the decision boundary is given as

$$f(c, b; Z) = c^T Z + b \quad (20)$$

where the feature vector “ Z ” for the i th dataset is defined as

$$Z_i = \begin{bmatrix} x_i & y_i & x_i^2 & y_i^2 & x_i y_i \end{bmatrix}^T, \quad i = 1, \dots, n \quad (21)$$

The coefficients c and b are data-dependent terms, which are obtained from the guidance dataset ($\mathcal{T}$). The goal is to find a separating function $f(c, b; Z)$ that maximally separates the controllable and uncontrollable regions while allowing for minimal misclassification. To establish a well-defined boundary, we introduce the classification constraint:

$$d_i(c^T Z_i + b) \geq 1 - \zeta_i, \quad i = 1, \dots, n \quad (22)$$

Here, ζ_i are slack variables that allow for minor classification errors, i.e., tolerating small violations of the margin. The constant 1 in the constraint $d_i(c^T Z_i + b) \geq 1 - \zeta_i$ ensures that the margin is normalized, thereby preventing arbitrary rescaling of c . Geometrically, this constraint ensures that data points lie on the correct side of the decision boundary and at least a distance of $(1/\|c\|)$ away from it when $\zeta_i = 0$. Therefore, the margin—the distance between the decision boundary and the closest data points—is inversely proportional to $\|c\|$. To maximize the margin, we equivalently minimize $(1/2)\|c\|^2$, resulting in the following convex optimization problem:

$$\begin{aligned}\text{Minimize}_{c,b} \quad & \frac{1}{2}\|c\|^2 + \gamma \sum_{i=1}^n \zeta_i \\ \text{Subject to} \quad & d_i(c^T Z_i + b) \geq 1 - \zeta_i, \quad i = 1, \dots, n \\ & \zeta_i \geq 0, \quad i = 1, \dots, n\end{aligned}\quad (23)$$

The scalar $\gamma > 0$ is a regularization parameter that controls the tradeoff between maximizing the margin (or minimizing $\|c\|^2$) and minimizing classification errors by penalizing large ζ_i . A larger γ emphasizes correct classification of training points, while a smaller γ allows more flexibility in margin violations to achieve better generalization.

The dual form of the optimization problem, Eq. (23), is given by

$$\begin{aligned}\text{Maximize}_{\alpha} \quad & \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n d_i d_j \alpha_i \alpha_j Z_i^T Z_j \\ \text{Subject to} \quad & 0 \leq \alpha_i \leq \gamma, \quad i = 1, \dots, n \\ & \sum_{i=1}^n \alpha_i d_i = 0\end{aligned}\quad (24)$$

where α_i are the Lagrange multipliers associated with the inequality constraints. This formulation is similar to the soft-margin classifier formulation in the context of the support vector machine [36]. The derivation of the dual form is included in Appendix A. The dual form in Eq. (24) is solved using the sequential minimal optimization (SMO) algorithm [37] implemented in MATLAB.

Figure 4 shows the solution of the optimization problem along with the support vectors associated with the solution. From a classification standpoint, the optimization yields a solution that minimizes misclassification error in the presence of overlapping classes. However, for planetary landing applications, even a single instance of false classification can lead to mission failure. Therefore, a zero-tolerance policy toward false classification is imperative. The zero-tolerance criterion cannot be achieved by simply tuning the hyperparameter (γ) in problem (23), as both regions overlap and the data are not linearly separable. To address this, we introduce

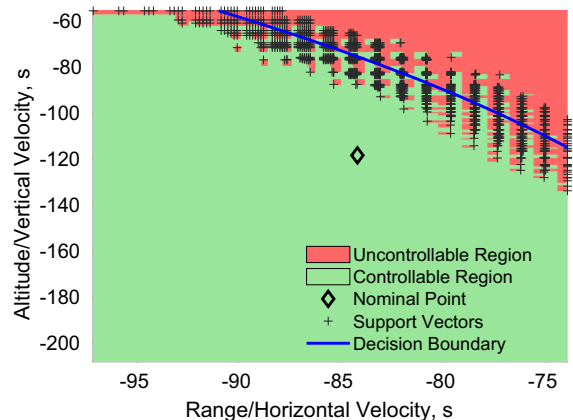


Fig. 4 Convex decision boundary.

an inner optimization layer designed to exclude uncontrollable data points from the decision boundary, thereby enforcing guaranteed safety.

Inner-Layer Optimization Problem: Any arbitrary shift of the decision boundary toward the controllable region leads to misclassifying controllable data as uncontrollable. This, in turn, reduces the size of the initial-state error ellipse for which pinpoint landing is achievable. Therefore, the decision boundary must be optimized to ensure zero false classification of uncontrollable data as controllable, while simultaneously minimizing the reduction in the landing ellipse caused by misclassification of controllable data.

We first rewrite Eq. (20) as a general conic given by

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0 \quad (25)$$

where A, B, C, D, E are obtained from coefficients (c, b) . We shift this conic from a nonzero origin to the $(0,0)$ coordinate by the following transformation:

$$x = X + h, \quad y = Y + k \quad (26)$$

which shifts the conic to the origin $(0,0)$, with (h, k) denoting the center of the original conic. Hence, Eq. (25) can be rewritten as

$$A_1X^2 + B_1XY + C_1Y^2 = 1 \quad (27)$$

where $A_1 = (-A/\lambda)$, $B_1 = -(B/\lambda)$, and $C_1 = -(C/\lambda)$. The parameters λ, h , and k are given by

$$\lambda = Ah^2 + Bhk + Ck^2 + Dh + Ek + F \quad (28)$$

$$h = \frac{2CD - BE}{B^2 - 4AC} \quad (29)$$

$$k = \frac{2AE - BD}{B^2 - 4AC} \quad (30)$$

Further, Eq. (27) is rotated to eliminate cross terms, resulting in

$$\bar{X}^2 + \bar{Y}^2 = 1 \quad (31)$$

where

$$\begin{bmatrix} \bar{X} \\ \bar{Y} \end{bmatrix} = R \begin{bmatrix} X \\ Y \end{bmatrix}, \quad R = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \quad (32)$$

The rotation angle θ is computed using the following expression:

$$\tan \theta = \frac{(C_1 - A_1) + \sqrt{B_1^2 + (C_1 - A_1)^2}}{B_1} \quad (33)$$

Substitution of Eqs. (32) and (26) in Eq. (31) yields

$$M_1(\cos \theta(x - h) + \sin \theta(y - k))^2 + M_2(-\sin \theta(x - h) + \cos \theta(y - k))^2 = 1 \quad (34)$$

$$M_1 = A_1 \cos^2 \theta + B_1 \cos \theta \sin \theta + C_1 \sin^2 \theta \quad (35)$$

$$M_2 = A_1 \sin^2 \theta - B_1 \cos \theta \sin \theta + C_1 \cos^2 \theta \quad (36)$$

The major and minor axes of the conic equation (34) are obtained using Eqs. (26) and (32) as

$$\cos \theta(x - h) + \sin \theta(y - k) = 0 \quad (37)$$

$$-\sin \theta(x - h) + \cos \theta(y - k) = 0 \quad (38)$$

Using Eq. (34), the decision boundary (20) can be written as

$$f(c, b; Z) = M_1(\cos \theta(x - h) + \sin \theta(y - k))^2 + M_2(-\sin \theta(x - h) + \cos \theta(y - k))^2 - 1 \quad (39)$$

Now, consider $\delta = [\delta_h, \delta_k, \delta_\theta]^T$, denoting the shift in the centroid and rotation of the conic equation (34). Then, the shifted conic parameterized by δ is given as

$$f'(\delta; Z) = M_1(\delta)(\cos \theta'(x - h') + \sin \theta'(y - k'))^2 + M_2(\delta)(-\sin \theta'(x - h') + \cos \theta'(y - k'))^2 - 1 \quad (40)$$

where $h' = h + \delta_h$, $k' = k + \delta_k$, and $\theta' = \theta + \delta_\theta$. Therefore, the inner-layer optimization problem is given as

$$\begin{aligned} &\text{Minimize}_\delta \quad \delta^T \delta \\ &\text{Subject to} \quad f'(\delta; Z^{(i)}) \leq \eta_i \quad i = 1, \dots, n \end{aligned} \quad (41)$$

The vector $\eta \in \mathbb{R}^n$ serves as a hyperparameter that governs the tradeoff between minimizing misclassification loss and preserving the size of the feasible error ellipse. Finally, combining problems (23) and (41) yields a zero-loss convex decision boundary represented as

$$f'(\delta; Z) = 0 \quad (42)$$

Figure 5 illustrates the zero-loss decision boundary resulting from the bilevel optimization. It is evident that the decision boundary satisfies all the constraints outlined in the formulation. During the mission, this decision boundary can be evaluated onboard in real time, and the sign of the evaluation alone is sufficient to determine the feasibility of pinpoint landing.

The resulting decision boundary is hyperbolic in nature. As a consequence, other branches of the hyperbola may incorrectly classify state vectors lying within them as controllable. Hence, evaluating the decision boundary at the current state vector provides only a necessary condition for determining controllability. To establish a sufficient condition, we incorporate an additional geometric constraint derived from the hyperbola's conjugate axis. Specifically, the state vector must also lie on the correct side of the conjugate axis to ensure convergence.

The expressions of the conjugate and transverse axes are obtained by evaluating Eqs. (37) and (38) for the zero-loss decision boundary [Eq. (42)]. The function corresponding to the conjugate axis is given as

$$g(\delta; Z) = x \cos \theta' + y \sin \theta' - (h' \cos \theta' + k' \sin \theta') \quad (43)$$

Figure 6 illustrates the transverse and conjugate axes of the conic decision boundary. Notably, any two state vectors that

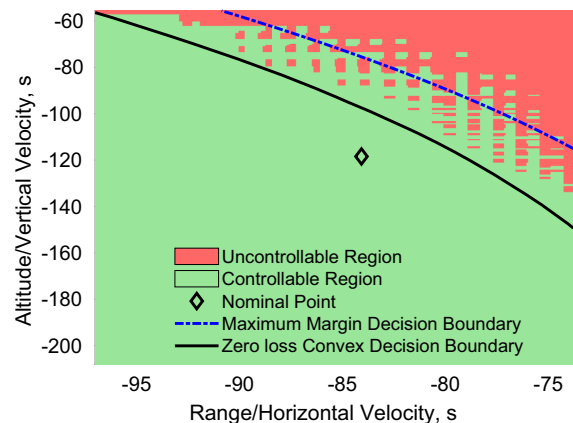


Fig. 5 Zero-loss convex decision boundary.

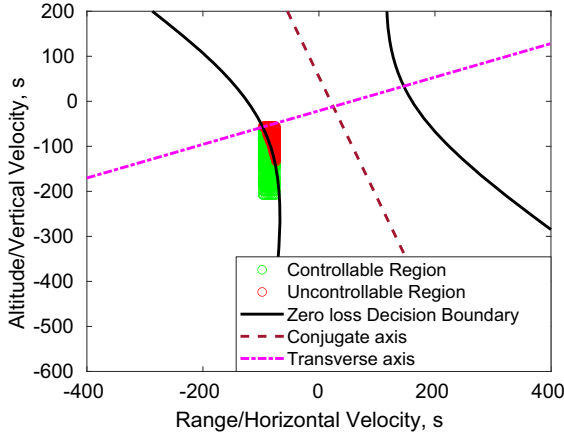


Fig. 6 Zero-loss convex decision boundary with conjugate and transverse axis.

yield the same sign when evaluated using the hyperbolic boundary function may still produce opposite signs when evaluated using the conjugate axis function in Eq. (43). This observation is crucial, as it highlights the potential ambiguity inherent in using the hyperbolic boundary alone for classification. To resolve this, we propose a two-step criterion that collectively serves as a necessary and sufficient condition for verifying controllability. First, the state vector is evaluated using the zero-loss decision boundary function in Eq. (40). A negative result immediately classifies the state as uncontrollable, whereas a positive result only suggests potential controllability. To confirm controllability, the state vector is then evaluated using the conjugate axis function in Eq. (43). A consistent sign in this second evaluation confirms that the state vector indeed lies within the controllable region.

B. Dynamic Retargeting

Once the decision boundary is characterized, the next step is to quantify the required retargeting if the current state vector is deemed uncontrollable. Typically, any point within the controllable region could serve as an alternative target. However, the retargeting is constrained to horizontal shifts within the dataset, as only the range component is adjustable. This requirement can be represented as

$$\mathbb{P}(x, y) = (x', y) \quad (44)$$

where $\mathbb{P}$ denotes a projection function that maps a point (x, y) to a new point (x', y) along the horizontal direction. While several candidate functions can satisfy this condition, they may result in arbitrary deviations from the nominal target. However, to maintain a near-pinpoint landing, it is essential to minimize the range shift. This is best achieved by using the decision boundary itself as the projection function. At the start of the braking phase, the state vector is first projected onto the conic decision boundary. Keeping the altitude, vertical velocity, and horizontal velocity constant, a new range value is determined by solving the quadratic equation derived. This yields two possible roots for the range. The correct root is chosen using the conjugate axis of the hyperbola. The complete algorithm is summarized in Algorithm 2.

VII. Simulations and Results

Extensive simulation experiments have been carried out using 6DoF simulation software to verify and validate the performance of the proposed guidance algorithm. The parameters of the problem are

Algorithm 2: Algorithm for real-time retargeting

v1: **Input:** State vector in MCMF at the start of fine braking phase, $(\mathbf{r}_0, \mathbf{v}_0)$, and nominal target at the end of phase, $(\mathbf{r}_f, \mathbf{v}_f)$

2: **Output:** New target state vector in MCMF, $(\mathbf{r}_{\text{new}}, \mathbf{v}_f)$ ($\mathbf{v}_f = \mathbf{0}$ at the fine braking end)

3: $\mathcal{X}_i = (H, S, w, v)^T \leftarrow (\mathbf{r}_0, \mathbf{v}_0)$,

4: $x \leftarrow \frac{S}{v}$, $y \leftarrow \frac{H}{w}$ and thereafter find Z_i from Eq. (21)

5: Evaluate $f'(Z)$ from Eq. (40)

6: **if** $f'(Z_i) < 0$ **then**

7: declare $\mathcal{X}_i \in \mathcal{T}_{nc}$ and goto Step 14

8: **else**

9: Evaluate Eq. (43)

10: **if** $g(Z_i) < 0$ **then**

11: declare $\mathcal{X}_i \in \mathcal{T}_c$

12: $\mathbf{r}_{\text{new}} \leftarrow \mathbf{r}_f$, and goto Step 19

13: **else**

14: Solve Eq. (42) for new x (denoted by x') for the current y

15: New range, $S' \leftarrow x'v$

16: $(\mathbf{r}_{\text{new}}, \mathbf{v}_f) \leftarrow (H, S', w, v)^T$

17: **end if**

18: **end if**

19: **return** $(\mathbf{r}_{\text{new}}, \mathbf{v}_f)$ and stop

$$\mathbf{g} = [-1.68 \quad 0 \quad 0]^T \text{ m/s}^2, \quad m_{\text{wet}} = 1050 \text{ kg}, \quad (45)$$

$$I_{sp} = 298.5 \text{ s}, \quad T_{\min} = 1480 \text{ N}, \quad T_{\max} = 3120 \text{ N}$$

Figure 7 shows the guidance control flow diagram for implementation. The simulation results presented in Fig. 8 show the time histories of position, velocity, and acceleration of the spacecraft, comparing the performance of the proposed guidance algorithm with fuel-optimal powered descent guidance (FOPDG). The lossless-convexification-based FOPDG, as discussed in [9], is implemented for comparison. The results in terms of fuel consumption and time of flight are presented in Table 2. As expected, the proposed data-driven polynomial guidance performs less optimally by 2.7 kg compared to the globally fuel-optimal solution, since the supervised-learning-based offline analysis identifies a fuel-optimal trajectory only within the class of functions parameterized by a cubic polynomial. It is well known that the true fuel-optimal solution exhibits a bang–bang structure in the acceleration profile, which cannot be captured by cubic functions. Nevertheless, the proposed polynomial guidance successfully identifies the fuel-optimal trajectory within this function class while adhering to the state and control constraints discussed earlier.

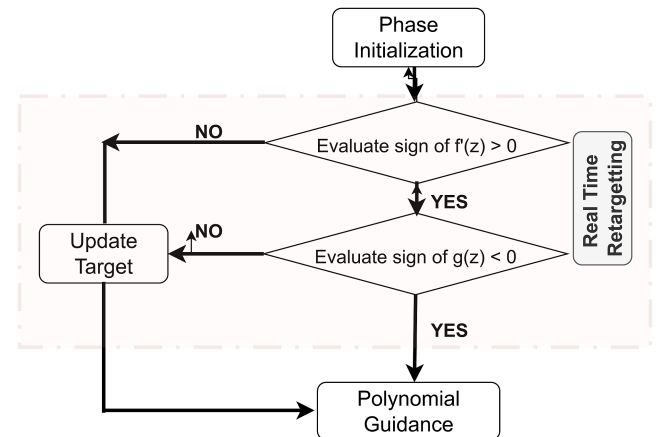


Fig. 7 Guidance control flow diagram.

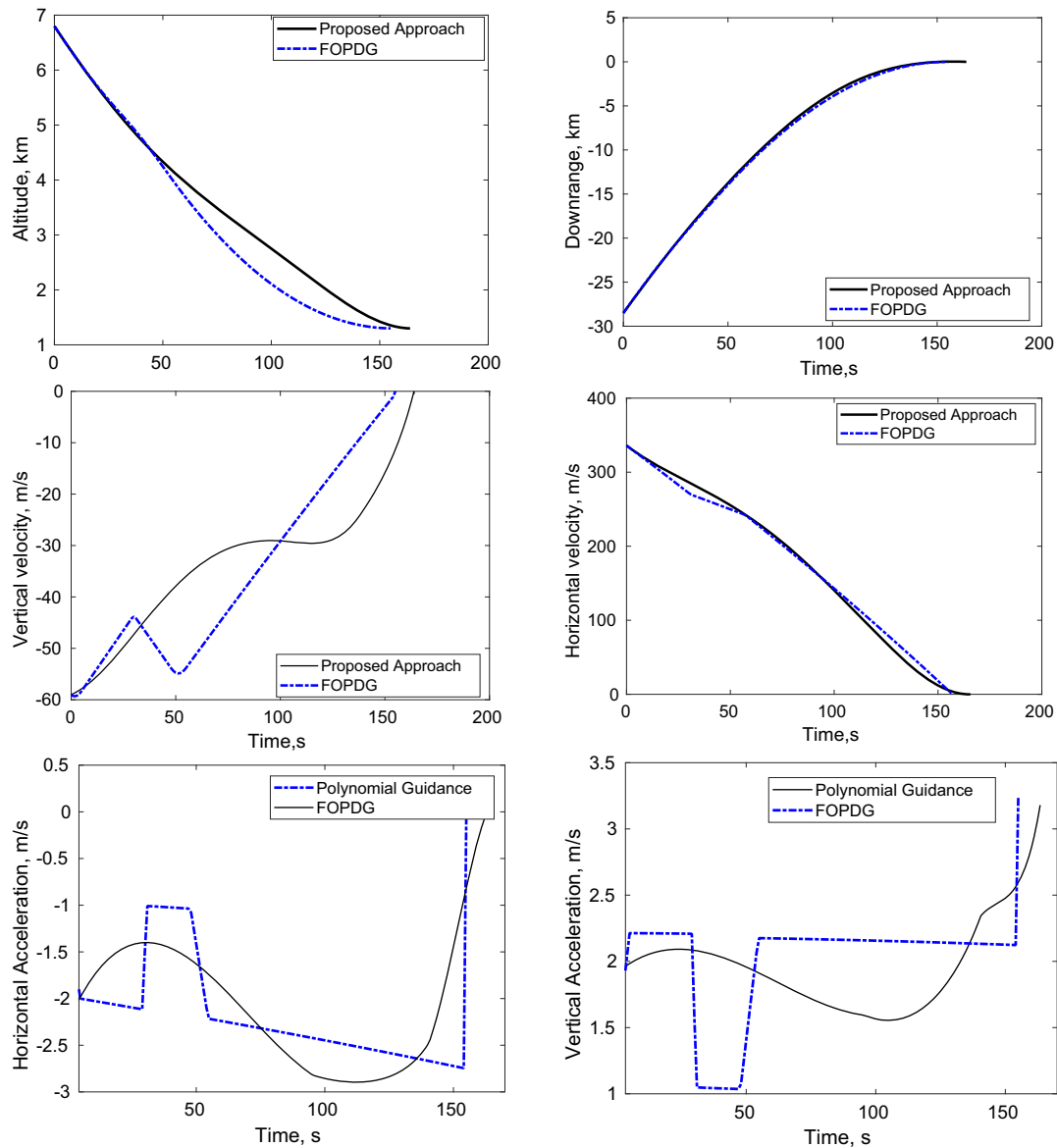


Fig. 8 Comparison of trajectory trace for proposed polynomial guidance and fuel optimal powered descent guidance: $r_0 = [6.8, -28.5, 0]^T$ km, $v_0 = [-59, 336, 0]^T$ m/s, $a_0 = [1.91, -2.26, 0]^T$ m/s², $r_T = [1.3, 0, 0]^T$ km, $v_T = [0, 0, 0]^T$ m/s, $a_T = [3.2, 0, 0]^T$ m/s².

Table 2 Performance comparison

Guidance algorithm	Fuel consumed, kg	Time of flight, s
Proposed polynomial guidance	154.5	164.4
Fuel optimal powered descent guidance	152.1	154.8

The simulation results presented in Fig. 9 demonstrate the robustness of the proposed guidance algorithm under off-nominal scenarios through autonomous real-time retargeting. We consider a scenario where the spacecraft starts the absolute navigation phase 2 km ahead and 1.5 km below the nominal state. Under such conditions, the guidance thrust demand remains fully saturated throughout the descent and fails to achieve the desired terminal altitude and velocity (shown in red). With the application of the real-time retargeting algorithm, a target shift of 3.027 km is computed autonomously, enabling the lander to reach the new target while satisfying all the constraints (shown in black).

A Monte Carlo analysis was carried out considering additive 3σ and beyond- 3σ dispersions in the nominal state of the spacecraft (as

specified in Table 1). The objective is to demonstrate the robustness of the proposed algorithm under state dispersions. Figure 10 presents a sample set of the range versus altitude, velocity, thrust, and cone angle w.r.t. vertical profiles. The blue curves indicate the cases where real-time retargeting is not needed. As the dispersion level increases beyond 3σ , pinpoint landing at the designated landing site is not feasible. This calls for retargeting to a different landing site, as shown by the green traces. The corresponding extended operating region is shown in Fig. 11. Out of all the data points of T_{nc} , 99.65% cases converge after performing the retargeting maneuver. The remaining 1.35% cases do not converge to the target due to violation of subtarget travel conditions during descent. For such cases, a salvage strategy has been worked out as part of the Chandrayaan-3 mission planning, which is beyond the scope of this paper.

VIII. Conclusions

In this paper, we proposed a robust and fuel-efficient polynomial guidance for the absolute navigation phase of the Chandrayaan-3 landing mission. The main contribution of this research is a robust decision-making algorithm that enables an online feasibility check for pinpoint landing and computes an alternative landing site if the

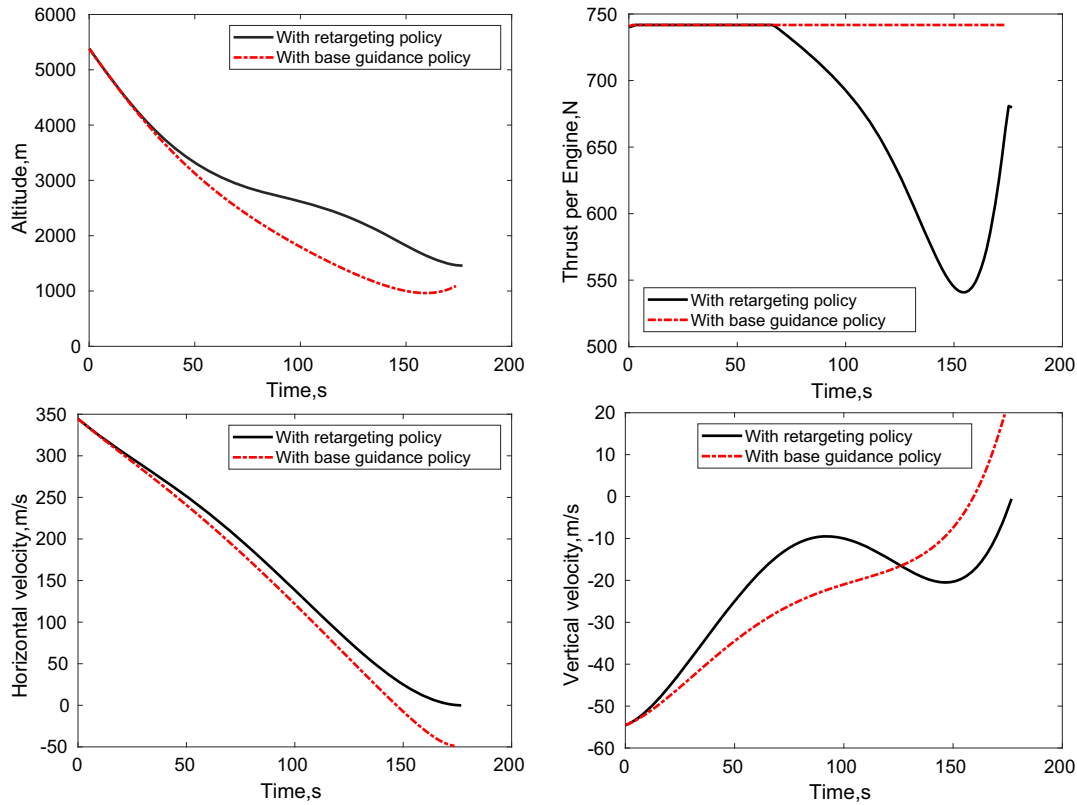


Fig. 9 Trajectory trace for real-time retargeting: $r_0 = [5.3, -26, 0]^T$ km, $v_0 = [-54, 344, 0]^T$ m/s, $a_0 = [1.91, -2.26, 0]^T$ m/s², $r_T = [1.3, 0, 0]^T$ km, $v_T = [0, 0, 0]^T$ m/s, $a_T = [3.2, 0, 0]^T$ m/s².

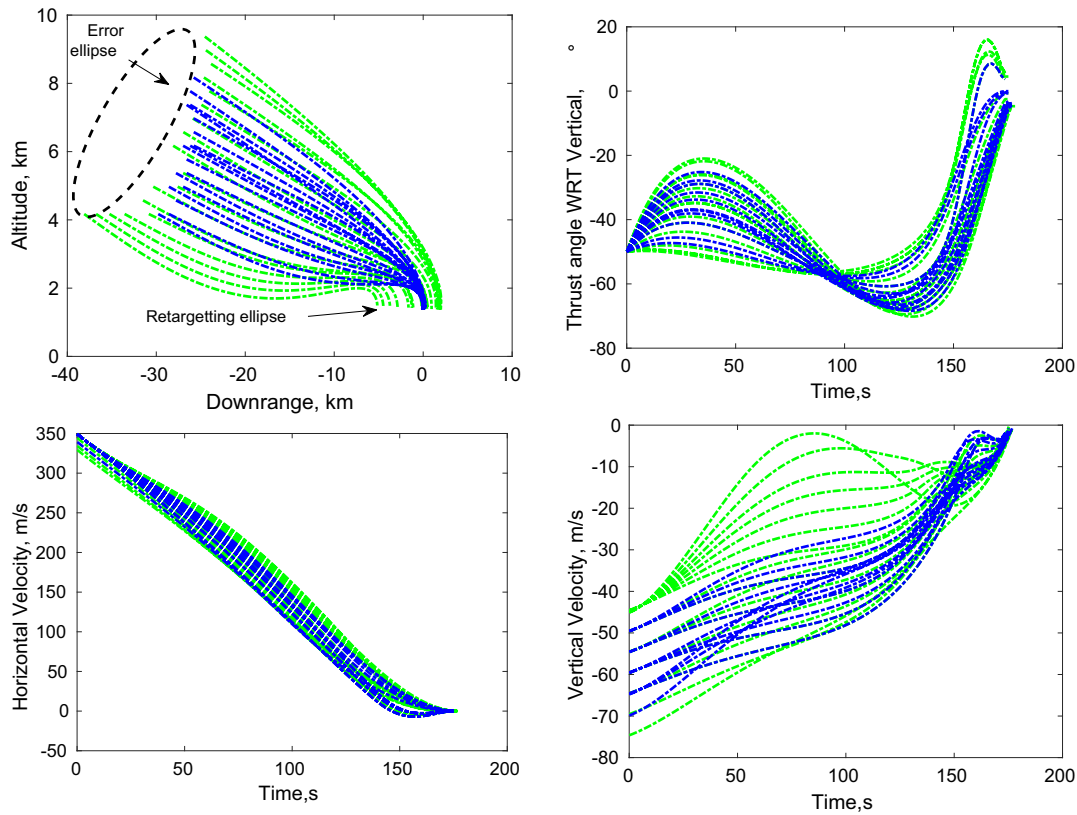


Fig. 10 Trajectory trace from Monte Carlo simulations.

pinpoint landing is infeasible. The algorithm considers state and control constraints in an offline data-driven analysis and therefore extends the operational region of the classical polynomial guidance. Another advantage of this algorithm is that it involves analytical

expressions for decision-making and target computation. This algorithm has been an integral part of the GNC design of the Chandrayaan-3 mission and played a pivotal role in ensuring the first soft landing near the lunar south pole.

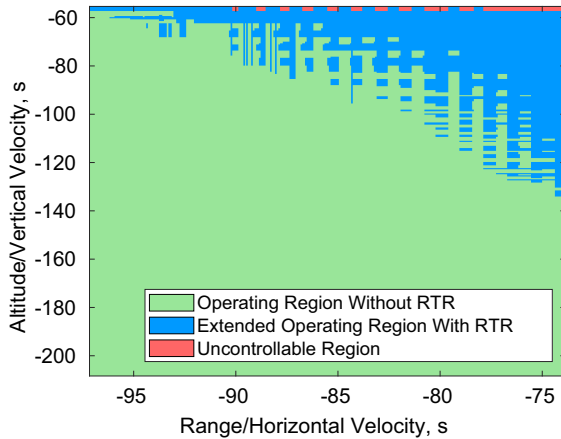


Fig. 11 Extended operating region from retargeting.

Appendix A: Derivation of SVM Dual Problem

The derivation of the SVM dual problem (24) for the SVM primal problem in Eq. (23) is given as follows [38]:

Start with the primal problem:

$$\begin{aligned} \text{Minimize}_{c,b} \quad & \frac{1}{2} \|c\|^2 + \gamma \sum_{i=1}^n \zeta_i \\ \text{Subject to} \quad & d_i(c^\top Z_i + b) \geq 1 - \zeta_i, \quad i = 1, \dots, n \\ & \zeta_i \geq 0, \quad i = 1, \dots, n \end{aligned}$$

A.1. Step 1: Construct the Lagrangian

Introduce dual variables: $\alpha_i \geq 0$ for the margin constraints and $r_i \geq 0$ for the slack-variable non-negativity.

The Lagrangian is

$$\begin{aligned} \mathcal{L}(c, b, \zeta, \alpha, r) = & \frac{1}{2} \|c\|^2 + \gamma \sum_{i=1}^n \zeta_i \\ & - \sum_{i=1}^n \alpha_i (d_i(c^\top Z_i + b) - 1 + \zeta_i) - \sum_{i=1}^n r_i \zeta_i \end{aligned}$$

A.2. Step 2: Take Derivatives and Apply KKT Conditions

With respect to c :

$$\frac{\partial \mathcal{L}}{\partial c} = c - \sum_{i=1}^n \alpha_i d_i Z_i = 0 \quad \Rightarrow \quad c = \sum_{i=1}^n \alpha_i d_i Z_i$$

With respect to b :

$$\frac{\partial \mathcal{L}}{\partial b} = - \sum_{i=1}^n \alpha_i d_i = 0 \quad \Rightarrow \quad \sum_{i=1}^n \alpha_i d_i = 0$$

With respect to ζ_i :

$$\frac{\partial \mathcal{L}}{\partial \zeta_i} = \gamma - \alpha_i - r_i = 0 \quad \Rightarrow \quad \alpha_i + r_i = \gamma$$

A.3. Step 3: Plug Back into the Lagrangian

Substitute the expressions for c , b , and the constraint on $\alpha_i + r_i$:

$$\begin{aligned} \mathcal{L} = & \frac{1}{2} \|c\|^2 + \sum_{i=1}^n \gamma \zeta_i - \sum_{i=1}^n \alpha_i (d_i(c^\top Z_i + b) - 1 + \zeta_i) - \sum_{i=1}^n r_i \zeta_i \\ = & \frac{1}{2} \left\| \sum_{i=1}^n \alpha_i d_i Z_i \right\|^2 - \sum_{i=1}^n \alpha_i d_i \left(\sum_{j=1}^n \alpha_j d_j Z_j^\top Z_i \right) + \sum_{i=1}^n \alpha_i \end{aligned}$$

On simplifying, we get

$$\mathcal{L}_D(\alpha) = -\frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j d_i d_j Z_i^\top Z_j + \sum_{i=1}^n \alpha_i$$

A.4. Step 4: Final Dual Problem

$$\begin{aligned} \text{Maximize}_{\alpha} \quad & \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j d_i d_j Z_i^\top Z_j \\ \text{Subject to} \quad & 0 \leq \alpha_i \leq \gamma, \quad i = 1, \dots, n \\ & \sum_{i=1}^n \alpha_i d_i = 0 \end{aligned}$$

Appendix B: 6DoF Simulation Framework

A high-fidelity 6DoF simulation framework was developed to support the design, validation, and performance assessment of the Chandrayaan-3 powered descent navigation, guidance, and control system. The simulator models the complete nonlinear translational and rotational motion of the lander, incorporating realistic environmental, propulsion, and sensor characteristics. It provides a closed-loop environment wherein the flight algorithms related to the GNC software are executed with simulated sensor feedback and actuator dynamics.

B.1. Dynamic Model

The lander dynamics are represented by the full nonlinear rigid-body equations of motion formulated in the body frame as

$$\begin{aligned} \dot{\mathbf{r}}_I &= \mathbf{v}_I, \\ m \dot{\mathbf{v}}_I &= \mathbf{R}_B^I \mathbf{F}_B + \mathbf{R}_B^I \mathbf{F}_s, \\ \dot{\mathbf{q}} &= \frac{1}{2} \boldsymbol{\Omega}(\boldsymbol{\omega}_B) \mathbf{q}, \\ \mathbf{J} \dot{\boldsymbol{\omega}}_B + \boldsymbol{\omega}_B \times (\mathbf{J} \boldsymbol{\omega}_B) &= \mathbf{M}_T + \mathbf{M}_s \end{aligned} \quad (\text{B1})$$

where m is the lander mass; $\mathbf{r}_I$ is the lander inertial position vector; $\mathbf{v}_I$ is the lander inertial velocity vector; $\mathbf{F}_T$ is the total thrust vector in the body frame; $\mathbf{F}_s$ and $\mathbf{M}_s$ are the equivalent slosh forces and torques in the body frame; $\boldsymbol{\omega}_B$ is the body rate with respect to the inertial frame expressed in the body frame; $\mathbf{R}_B^I$ is the rotation matrix of transformation from the body frame to the inertial frame; $\mathbf{J}$ is the inertia matrix; $\mathbf{q}$ is the attitude quaternion; and $\mathbf{M}_T$ is the control moment from engines and attitude thrusters.

Body acceleration is sensed by the accelerometer and transformed to the inertial frame. The transformed body acceleration is then used for navigation via the accelerometer simulation model.

The lunar gravitational field is modeled using a spherical harmonic expansion up to order 30 derived from the Gravity Recovery and Interior Laboratory (GRAIL) mission data:

$$\begin{aligned} U(r, \lambda, \phi) = & \frac{\mu}{r} \sum_{n=0}^{30} \left(\frac{R_m}{r} \right)^n \sum_{m=0}^n \bar{P}_{nm}(\sin \phi) [\bar{C}_{nm} \cos(m\lambda) \\ & + \bar{S}_{nm} \sin(m\lambda)] \end{aligned} \quad (\text{B2})$$

where μ is the lunar gravitational parameter, R_m is the mean lunar radius, $\bar{C}_{nm}$ and $\bar{S}_{nm}$ are normalized spherical harmonic coefficients, and $\bar{P}_{nm}$ are associated Legendre polynomials. The gravitational acceleration is obtained as $\mathbf{a}_g = -\nabla U$.

The slosh dynamics consist of two segments: analogous pendulum dynamics and disturbance torque emulating the effects of sloshing on the spacecraft. Each propellant tank is represented as a sloshing mass attached to the tank wall by a pendulum of given length. The dynamic nature of the propellant participating in slosh is

captured in terms of time-varying analogous pendulum parameters. The detailed modeling aspects are discussed in [39].

The propulsion subsystem comprises throttleable main engines and multiple attitude thrusters. The main engine thrust is represented by a first-order empirical response model derived from test data:

$$\dot{T}_c = \frac{1}{\tau_T}(T_{\text{cmd}} - T_c) \quad (\text{B3})$$

where T_c is the chamber thrust, T_{cmd} is the commanded thrust, and τ_T is the engine time constant. The net thrust and moment acting on the lander are computed as

$$\mathbf{F}_T = \sum_i T_i \mathbf{u}_i \quad (\text{B4})$$

$$\mathbf{M}_T = \sum_i \mathbf{r}_i \times (T_i \mathbf{u}_i) \quad (\text{B5})$$

where T_i , $\mathbf{u}_i$, and $\mathbf{r}_i$ are the thrust magnitude, direction, and position vector of the i th engine, respectively.

Sensor models emulate the behavior of the onboard navigation instruments, including biases, scale factors, noise, and latency effects. The Inertial Measurement Unit (IMU) is modeled as

$$\tilde{\mathbf{a}} = \mathbf{a}_B + \mathbf{b}_a + \mathbf{n}_a \quad (\text{B6})$$

$$\tilde{\boldsymbol{\omega}} = \boldsymbol{\omega}_B + \mathbf{b}_g + \mathbf{n}_g \quad (\text{B7})$$

where tildes denote measured quantities, $\mathbf{b}_a$ and $\mathbf{b}_g$ are biases, and $\mathbf{n}_a$ and $\mathbf{n}_g$ are Gaussian white noise terms. Radar altimeter and Doppler velocimeter models include realistic line-of-sight constraints, delay, and dropout logic. Star tracker measurements are generated at discrete update rates with field-of-view and noise limitations.

B.2. Terrain and Environment Modeling

A digital elevation model (DEM) of the landing site, derived from Chandrayaan-2 orbiter high-resolution camera data (OHRC), is integrated to represent local topography. The DEM supports terrain-relative navigation (TRN) and absolute-sensor line-of-sight simulation.

B.3. Onboard Software Integration and Frame Transformations

The Chandrayaan-3 onboard GNC software is integrated within the 6DoF simulation loop. The software operates on simulated sensor inputs and produces control commands to the propulsion and attitude control models at flight-representative update rates. All coordinate-frame transformations are implemented consistently with the flight architecture:

$$\text{MCI} \leftrightarrow \text{MCMF} \leftrightarrow \text{Local (NUE)}$$

where MCI denotes the moon-centered inertial frame, MCMF the moon-centered-moon-fixed frame, and NUE the local north-up-east landing frame.

B.4. Simulation Validation and Flight Correlation

The developed simulator was validated at both subsystem and system levels before flight use. Each subsystem model was benchmarked against analytical and experimental data. The throttleable engine model reproduced measured transient responses within 3% RMS deviation, while the slosh model matched the dominant modal frequencies within 5%. IMU and altimeter sensor noise characteristics were tuned to match qualification data via Allan variance analysis.

The integrated simulator was executed with the onboard GNC software in the loop. Monte Carlo campaigns were performed with dispersions in the initial state vector, mass, thrust, sensor bias,

drift, and propulsion parameters. Each realization was propagated from 30 km altitude to touchdown. Landing dispersions, terminal attitude errors, and propellant consumption were recorded. For 1000 Monte Carlo runs, the $1\text{-}\sigma$ horizontal landing dispersion was within 200 m and the terminal attitude error within 0.5 deg, meeting Chandrayaan-3 design criteria. Postlanding analysis compared flight telemetry with simulation predictions. Let $\mathbf{x}_{\text{sim}}(t)$ and $\mathbf{x}_{\text{flight}}(t)$ denote simulated and measured state vectors. The normalized trajectory error norm was computed as

$$\varepsilon(t) = \frac{\|\mathbf{x}_{\text{sim}}(t) - \mathbf{x}_{\text{flight}}(t)\|}{\|\mathbf{x}_{\text{flight}}(t)\|} \quad (\text{B8})$$

Throughout descent, $\varepsilon(t)$ remained below 2% for position and below 1 deg for attitude. Simulated throttle histories closely matched telemetered engine data, validating the propulsion model fidelity. Minor deviations near terminal descent were attributed to local terrain slope and radar latency, within modeled uncertainty bounds.

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